3. Physics of Sound

Introduction

In the last chapter, the relationship between sound-pressure level and sound-power level was discussed. The relationship was given as:

$$SPL = SWL - 20 \text{ Log } r - 0.5 \text{ dB} + T + 10 \text{ Log } Q$$

where SPL denotes sound-pressure level (reference: $0.00002 N/m^2$), SWL denotes sound-power level (reference: 10^{-12} watts), r denotes distance from the sound source in feet, T denotes the correction for atmospheric conditions in dB, and Q denotes the directivity factor of the sound source.

This chapter discusses sound in a room. The relationship between sound-pressure level and sound-power level will be modified to account for the behavior of sound in a room. Also discussed in this chapter is how sound travels from one room to another. This will include the concepts of transmission loss and noise reduction.

Sound in a Room

Sound in a free-field, a condition where sound from a sound source is free to radiate in all directions, was presented in the last chapter. Sound in a free-field can be graphically represented by Figures 4.3.1 and 4.3.2.

Figure 4.3.1

Sound in a free field.

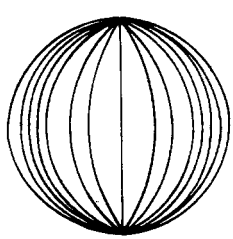
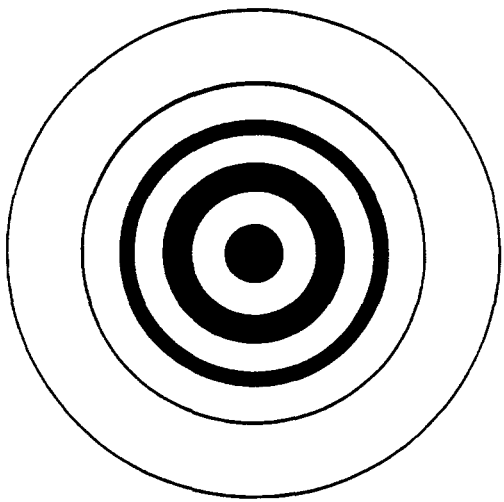


Figure 4.3.2

Sound pressure or intensity in a free field.



But what would happen if walls were put around the sound source? By putting the sound source in a room, what is going to happen when the sound reaches a wall? Naturally, the sound will be reflected off the walls--the sound will generate a reflected wave off the wall. (See Figure 4.3.3.) Actually, there would be multiple reflections; the sound waves will bounce off the walls repeatedly. (See Figure 4.3.4.)

From Figure 4.3.4, it can be seen that there are three fields of sound. There is sound that is in the near field--the field near the sound source. There is a field that acts like the inverse-square field in a free-field condition--the field that is some distance away from the sound source. In the inverse-square field, the sound of the sound source predominates, rather than the sound bouncing off the walls. And there is the sound field near the walls. This is the reverberant sound field where sound is reflected off the walls, and the reflected sound predominates rather than the sound from the sound source.

In Figure 4.3.4, sound intensity is used to show the three fields. However, sound pressure could have been used since sound intensity and sound pressure are related.

When the sound source is in a room, the sound pressure (or sound-pressure level) in the near field cannot be predicted. However, in the inverse-square field, the sound-pressure level or the sound-intensity level can be predicted if the sound-power level is known:

Figure 4.3.3

Sound in a room.

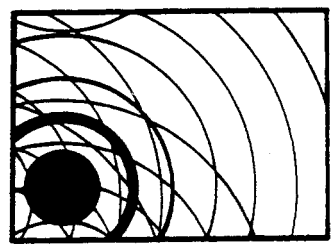
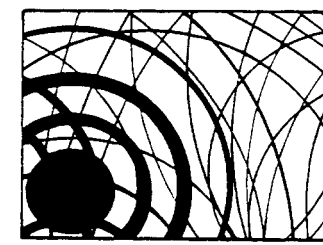
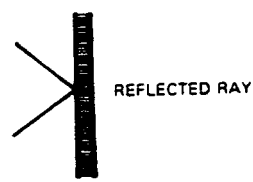


Figure 4.3.4

Multiple reflections.





$$\text{Sound intensity} = \frac{\text{Sound power}}{4\pi r^2}$$

where r denotes the distance from the sound source.

The sound pressure predicted is dependent upon the distance from the sound source. In the reverberant field, as can be seen from Figure 4.3.4, the sound pressure (and sound intensity) will eventually become constant near the walls and may have nothing to do with the distance and direction of the sound source; i.e., the sound pressure in the reverberant field will be independent of distance from the sound source. A method for predicting the sound pressure in the reverberant field is presented later in this chapter.

An important concept when considering sound in a room is the critical distance. The critical distance is the distance at which the noise level (sound pressure or intensity) changes from inverse-square-law behavior to reverberant field behavior. An expression for computing the critical distance will be given later in this chapter. There is a way of determining precisely the location where the inverse-square law no longer holds, and the reverberant field behavior of sound becomes the important sound. One can roughly estimate the critical distance by ear. This can be done by focusing your attention on a predominately identifiable noise source. Walk away from the source machine until the level appears relatively constant to your ears. Then walk rapidly toward the source, stopping when you are first aware of the increase in level. Repeat this procedure until you have localized the point at which the

direct sound begins to predominate, at the critical distance position. Then measure from there to the sound; this gives the critical distance.

Absorption

What factors will affect sound in the reverberant field? An obvious answer is the composition of the walls in the room. If the walls of the room are constructed of a hard material, the sound will reflect. On the other hand, if the walls of the room are constructed of a soft surface, sound will be absorbed rather than reflected. A soft surface is a surface that is made of porous material. This porous material has interconnecting air spaces that allow the sound to be absorbed. As the sound enters these interconnected air spaces, the sound energy is transformed into heat energy. (At normal sound intensities, this heat buildup is insignificant.) Thus, as the wall becomes more absorbent, there is less reflected sound; thus, there is less reverberant sound in the reverberant sound field.

The ability of a surface to absorb sound is given by the absorption coefficient, usually denoted by alpha (α). A surface that has an absorption coefficient equal to 1 is a perfectly absorbing medium, such as open space. Open space will not reflect any sound at all. A hard surface, such as marble, has an absorption coefficient equal to 0; that is, no sound is absorbed at all, and it is reflected back into the room.

The absorption coefficient of a given material is not the same for all frequencies. Given materials, such as marble, wood, plaster, etc., all have absorption coefficients; but these coefficients are different for different frequencies. There are tables available that give the absorption coefficient of a given material at a given frequency. Figure 4.3.5 shows the absorption coefficients of various materials at various frequencies.

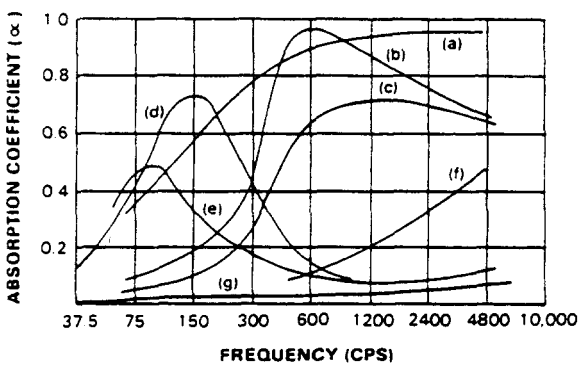
Most walls in industrial settings are not composed of just one material. For example, brick walls may have glass windows. Thus, what is needed is a system to compute the average absorption coefficient of a wall that is made of different kinds of materials. If a wall is made of glass and brick, what is the absorption coefficient of this wall? Before this question can be answered, the frequency of concern must be known since the absorption characteristics of the wall are dependent upon the frequency of the sound. Also, the area of the wall that is made of brick and the area of the wall that is made of glass must be known. Once this information is given, the average absorption coefficient of the wall can be computed using the following expression:

$$\alpha_{avg} = \frac{\sum_{i=1}^{n} S_i \alpha_i}{\sum_{i=1}^{n} S_i}$$

where α_{avg} denotes the average absorption coefficient of the wall constructed of combined materials; S_i denotes the surface area of the ith building material; and α_i denotes the absorption coefficient of the ith building material. (α_i would be obtained from a figure or table, such as Figure 4.3.5, given a specific frequency.)

Figure 4.3.5

Absorption coefficient.



The absorption coefficient of various materials (a reverberant sound field is assumed): (a) 4-inch glass wool blanket (6 lb/ft^3 density), against wall--no facing; (b) average for perforated acoustic tiles cemented directly to a hard backing and 1 inch thick; (c) average for perforated acoustic tiles cemented to a hard backing and 1/2-inch thick; (d) 1-inch mineral wool blanket (Aerocor--2 lb/ft^3) covered by perforated facing (12 percent open area) and separated from hard backing by 1-inch air space; (e) 3/8-inch plywood panel with 4-inch air space, one cross brace, and 2 inches of mineral wool at wall; (f) heavy carpet on concrete, (g) unpainted brick wall.

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For example, a room is 8 feet high, 10 feet wide, and 12 feet long. The ceiling is composed of acoustical tile that has an absorption coefficient of 0.95 at the frequency of concern; the floor is carpeted and has an absorption coefficient of 0.37 at the frequency of concern; and the walls are all made of brick which has an absorption coefficient of 0.03 at the frequency of concern. What is the average absorption coefficient of the room?

$$\alpha_{avg} = \frac{\sum_{i=1}^{n} S_i\alpha_i}{\sum_{i=1}^{n} S_i}$$

$$= \frac{(10 \times 12)0.95 + (10 \times 12)0.37 + 2(10 \times 8)0.03 + 2(12 \times 8)0.03}{2(10 \times 12) + 2(10 \times 8) + 2(12 \times 8)}$$

$$= \frac{168.96}{592}$$

$$= 0.285$$

But 0.285 what? What units are attached to the absorption coefficient? The absorption coefficient is commonly expressed in sabins. One sabin is equivalent to one square foot of a perfectly absorbent surface. The closer the absorption coefficient is to 1, the more absorbent that surface is. The closer the absorption coefficient is to 0, the harder the surface is, and more of the sound will be reflected.

Room Constant

Along with the absorption coefficient of the surfaces, the room constant is also of concern. The room constant is a measure of the ability of a room to absorb sound. The room constant, R, is a value based on the summation of all areas and is given by the following expression:

$$R = \frac{\alpha_{avg} \sum_{i=1}^{n} S_i}{1 - \alpha_{avg}} \quad \text{or} \quad R = \frac{\sum_{i=1}^{n} S_i\alpha_i}{1 - \alpha_{avg}}$$

where R denotes the room constant, α_{avg} denotes the average absorption coefficient of a room, and α_i denotes the absorption coefficient of the ith surface.

Given this expression, what is the room constant of a room that has the following characteristics? 8 feet by 12 feet by 10 feet; the ceiling has an absorption coefficient of 0.95 sabins; the floor has an absorption coefficient of 0.37 sabins; and the walls are brick and have an absorption coefficient of 0.03 sabins. To find the room constant, first the absorption coefficient of the room must be computed using the following expression:

$$\alpha = \frac{\sum_{i=1}^{n} S_i ft^2 \, \alpha_i sabins}{\sum S_i ft^2}$$

$$= \frac{168.96 \text{ ft}^2 \text{ sabins}}{592 \text{ ft}^2}$$

$$= 0.285 \text{ sabins}$$

Given the average absorption coefficient of the room, the room constant can be computed:

$$R = \frac{\alpha_{avg} \sum_{i=1}^{n} S_i}{1 - \alpha_{avg}}$$

$$= \frac{(0.285 \text{ sabins})\ (592 \text{ ft}^2)}{1 - 0.285 \text{ sabins}}$$

Note that the room constant would be in units of ft^2.

$$= 235.97 \text{ ft}^2$$

Factors Affecting the Reverberant Field. From the last two sections, it should be clear that sound pressure in the reverberant field depends upon the room dimensions, the number and type of objects in the room, the placement of the objects in the room, and the absorption characteristics of the room and the objects in the room. In the reverberant fields where a high percentage of reflected sound is present, the sound-pressure levels may be essentially independent of direction and distance to the sound source.

Room Constant and Sound Pressure

The room constant will affect the sound-pressure level in various parts of the room. If the walls were perfectly absorbent and no sound were reflected, the sound-pressure level could be computed as follows:

$$SPL_{(r)} = SPL_{(1)} - 20 \text{ Log } (r/r_1)$$

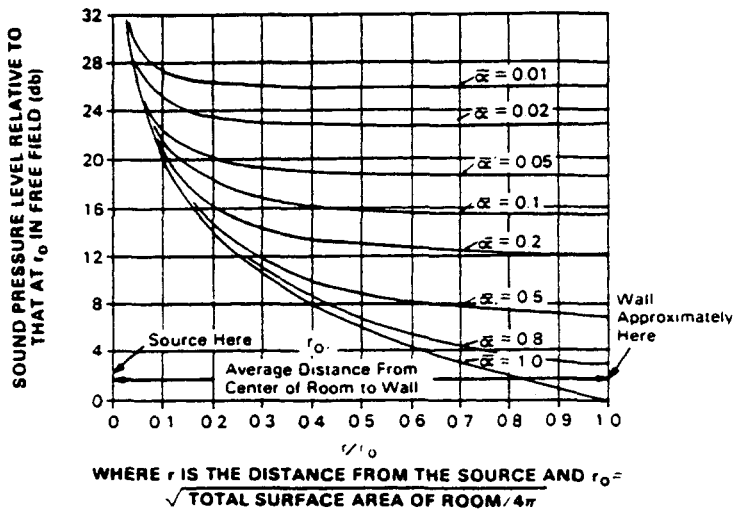
Where $SPL_{(r)}$ denotes the sound-pressure level at distance r, $SPL_{(1)}$ denotes a known sound-pressure level at some distance r_1, and r and r_1 denote the distances corresponding to $SPL_{(r)}$ and $SPL_{(1)}$ respectively.

If the sound-pressure level is known at a certain position, then the sound-pressure level can be calculated at any other position in the room. Where the walls are perfectly absorbent, sound is going to behave as it would in a free-field condition. Where the walls are less than perfectly absorbent and the sound is reflected off the walls, the sound-pressure levels in various parts of the room will not follow the relationship given in the above expression. Given the room constant, however, the relative sound-pressure level in the reverberant field can be calculated. Calculations from rooms having various absorption coefficients, α_{avg}, are shown in Figure 4.3.6.

[Note: α_{avg} is a function of the room constant: $\alpha_{avg} = R/(\Sigma S + R)$.]

Figure 4.3.6

Relative sound pressure level.



The horizontal axis is the distance, r, from the source to some desired point divided by r_o, the radius of a sphere whose surface area is the same as that of the room; i.e.,

$$r_o = \sqrt{\text{surface area}/4\pi}$$

For example, if the room is 8 feet by 12 feet x 10 feet, then r_o can be computed as

$$r_o = \sqrt{\text{surface area}/4\pi}$$

$$= \sqrt{2(10\text{ft} \times 8\text{ft}) + 2(8\text{ft} \times 12\text{ft}) + 2(10\text{ft} \times 12\text{ft})/4(3.1416)}$$

$$= \sqrt{592/12.56}$$

$$= 6.865 \text{ ft}$$

r_o represents the average distance from the center of the room to the walls.

The vertical axis shows the increase in sound-pressure level that exists at any point in the room over the sound-pressure level that would exist at r_o with the source radiating into the free space; e.g., perfectly absorbing walls. For example, if α_{avg} = 0.2 and the room has the following dimensions, 8 ft by 12 ft by 10 ft, then at a distance of 3.865 feet from the source, the sound-pressure level would be 14 dB higher than the sound-pressure level that would exist at 6.865 feet with the sound source radiating into a free space.

From Figure 4.3.6, it can be seen that in an acoustically treated room (with less than perfectly absorbing walls) as the distance from the sound source increases, the sound-pressure level will decrease for a while; but a point will be reached where there is little or no decrease. Specifically, a point will be reached where the sound-pressure level will level off. The sound-pressure level will not continue to decrease because of the reverberant buildup near the walls. The curves in Figure 4.3.6 are appropriate only where the observer and the sound source are at least one-half wavelength away from the walls. Thus, if the room is small or the frequency low, there will be positions in the room where Figure 4.3.6 will give incorrect results.

Given that the room constant affects the relative sound-pressure level at various distances from the sound source, it is obvious that the room constant also affects the relationship between the sound-power level and the sound-pressure level. The room constant affects the relationship between the sound-pressure level and the sound-power level in the following manner. (See: Industrial Noise Manual.)

$$SPL = SWL + 10 \text{ Log } [(Q/4\pi r^2) + (4/R)] + 10.5 + T$$

Assume a chipping hammer that radiates one watt of power in the center of a room. The directivity factor of the machine is 1; that is Q = 1. The average distance from the wall to the sound source is 6.87 feet, and the room constant is 229.39 ft^2. There is no correction for atmospheric temperature or pressure; i.e., T = 0. What is the sound-pressure level at the wall?

First, the sound-power level must be computed since the problem states only sound power and not sound-power level. Recall from the last chapter that the sound-power level can be computed using the following expression:

$$SWL = \text{Log } W + 120 \text{ (reference: } 10^{-12} \text{ watts)}$$

Substituting the sound power in this expression results in

$$SWL = \text{Log } 1 + 120$$

$$= 120 \text{ dB (reference: } 10^{-12} \text{ watts)}$$

Now the sound-power level can be substituted in the expression that shows the relationship between sound-power level and sound-pressure level. This results in the following:

$$SPL = SWL + 10 \text{ Log } [(Q/4\pi r^2) + (4/R)] + 10.5 + T$$

$$= 120 \text{ dB} + 10 \text{ Log } [(1/(4 \times 3.14 \times 6.87)) + (4/229.39)] + 10.5 + 0$$

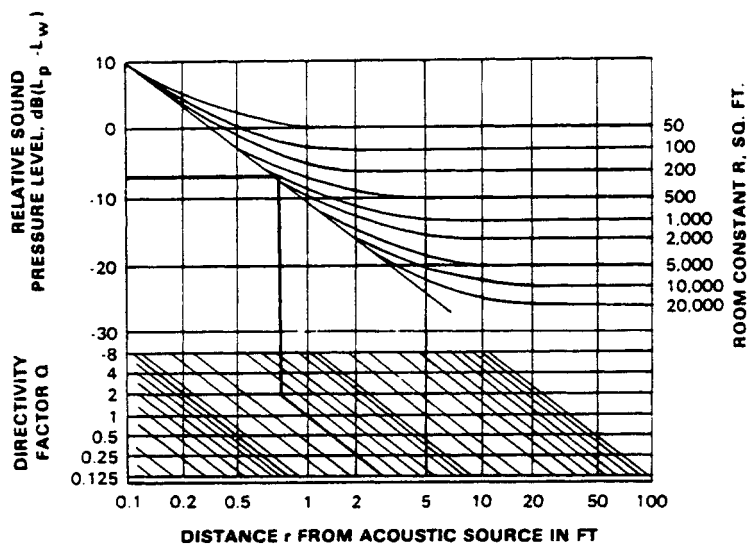
$$= 113.316 \text{ dB (reference: } 0.00002 \text{ N/m}^2)$$

Thus, it can be seen that the sound-pressure level at the wall would be 6.68 dB less than the sound-power level (or 113.316 dB). Note that this sound-pressure level takes into consideration the reverberant field because the expression used considered the room constant (R).

Rather than using the above expression to compute the sound pressure relative to the sound-power level, the relative sound pressure can be found by using published curves. (See Figure 4.3.7.) The figure is used by entering the curves at the horizontal axis at the desired distance, r; following the diagonal lines to a given Q; then up to the appropriate room-constant curve and across to reveal the sound-pressure level relative to the sound-power level (SPL - SWL). In the example given above, enter the curves at r = 6.87 feet; follow the diagonal line to Q = 1; then move up to a room constant of 229.39 feet^2 and across to read the sound-pressure level relative to sound-power level (about 7 dB); i.e., the sound-pressure level would be about 7 dB less than the sound-power level.

Figure 4.3.7

Sound-pressure level in a room relative to sound-power level.



Critical Distance

In previous sections of this chapter, it was noted that there are three sound fields in a room; the near field, the inverse-square-law field, and the reverberant field. It was also noted that in the inverse-square-law field the relationship between sound-power level and sound-pressure level is given as

$$SPL = SWL - 20 \text{ Log } r + 10 \text{ Log } Q - 0.5dB + T \quad (4.3.1)$$

where r is given in feet, SPL reference: 0.00002 N/m^2, and SWL reference: 10^{-12} watts.

In the reverberant field, the relationship between sound-pressure level and sound-power level can be expressed as

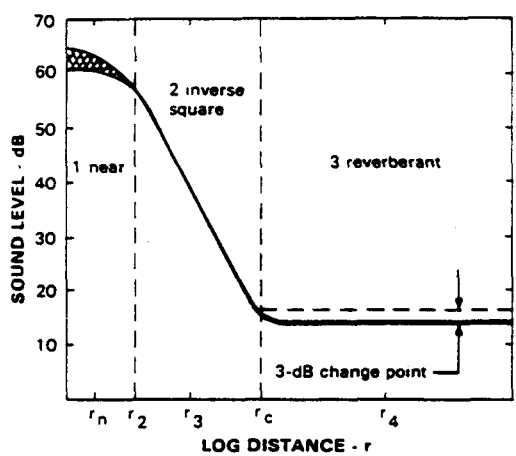
$$SPL = SWL + 10 \text{ Log } [(Q/4\pi r^2) + (4/R)] + 10.5 + T \qquad (4.3.2)$$

where r is given in feet, SPL reference: 0.00002 N/m^2, SWL reference: 10^{-12} watts, and R is given in square feet.

Pictorially, the three fields can be represented as in Figure 4.3.8.

Figure 4.3.8

Sound field in a room.



Of particular importance is the critical distance, r_c. The critical distance is the point at which the sound behavior changes from inverse-square-law behavior to reverberant field behavior. The junction between regions 2 and 3 in Figure 4.3.8 is the critical distance.

From expression (2) in this section, it is seen that the inverse-square-law and reverberant fields are equal. (See: Industrial Noise Control Manual, p. 59.)

$$\frac{Q}{4\pi r_c^2} = \frac{4}{R}$$

where r_c, denotes the critical distance in feet, Q denotes the directivity factor, and R denotes the room constant in square feet.

Solving the above identify for r_c results in

$$r_c = \sqrt{QR/16\pi} \qquad (4.3.3)$$

It is often convenient to use expression (4.3.3) in terms of α_{avg}, the average room coefficient. This can be done by recalling.

$$R = \Sigma S\alpha_{avg}/(1-\alpha_{avg}) \qquad (4.3.4)$$

Substituting expression (4.3.4) in expression (4.3.3) results in

$$r_c = \sqrt{[Q(\Sigma\alpha_{avg})/(1-\alpha_{avg})]/16\pi}$$

$$r_c = \sqrt{(Q\Sigma S\alpha_{avg})/16\pi(1-\alpha_{avg})}$$

$$r_c = 0.14\sqrt{(Q\Sigma S\alpha_{avg})/(1-\alpha)_{avg}}$$

If a room is 8 ft x 10 ft x 12 ft and has an average absorption coefficient of 0.28 sabins and a sound source with a Q factor of 1, where does the inverse-square-law field end and the reverberant field begin? (In other words, calculate r_c.)

$$r_c = 0.14\sqrt{1\ [2(8 \times 10 + 2(10 \times 12) + 2(8 \times 12)]\ 0.28/(1 - 0.28)}$$

$$= 0.14\sqrt{592(0.28)/0.72}$$

$$= 0.14\sqrt{165.78/0.72}$$

$$= 2.12 \text{ feet}$$

The critical distance is important because most workers are between the sound source and the critical distance point. The critical distance is also important because it can be shown that making the room more absorbent will not significantly affect the sound-pressure level at the critical distance. The problem below illustrates this point. In the problem, the sound-pressure level will be calculated in three locations; somewhere in the inverse-square-law field, at the critical distance, and somewhere in the reverberant field. Then the room will be made more absorbent to see how the sound pressure is affected in these three locations. For purposes of simplicity, the problem will be concerned only with sound having a frequency of 500 Hz.

Problem. The room is 8 feet high, 12 feet wide, and 10 feet long. The ceiling has an absorption coefficient of 0.95 sabins at 500 Hz; the floor has an absorption coefficient of 0.37 sabins at 500 Hz; and the walls have an absorption coefficient of 0.03 sabins at 500 Hz (brick). The sound source is a compressor for which the manufacturer reports a sound power of 110 dB (reference: 10^{-12} watts) and a directivity factor of 2.0. Assume normal atmospheric conditions; i.e., T = 0.

Solution. First, compute the critical distance, the point where the inverse square law ends and the reverberant field begins. Recall

$$r_c = \sqrt{QR/16\pi}$$

where

$$R = \Sigma S_i\alpha_i/(1 - \alpha_{avg})$$

and where $\alpha_{avg} = \Sigma S_i\alpha_i/\Sigma S_i$

then

$$\alpha_{avg} = \frac{(12\text{ ft} \times 10\text{ ft})\ 0.95\text{ sabins} + (12\text{ ft} \times 10\text{ ft})\ 0.37\text{ sabins} + 2(12\text{ ft} \times 8\text{ ft})\ 0.03\text{ sabins} + 2(10\text{ ft} \times 8\text{ ft})\ 0.03\text{ sabins}}{2(12\text{ ft} \times 10\text{ ft}) + 2(12\text{ ft} \times 8\text{ ft}) + 2(10\text{ ft} \times 8\text{ ft})}$$

$$= \frac{114\text{ ft}^2\text{sabins} + 44.4\text{ ft}^2\text{sabins} + 5.76\text{ ft}^2\text{sabins} + 4.8\text{ ft}^2\text{sabins}}{592\text{ ft}^2}$$

$$= \frac{168.96\text{ ft}^2\text{sabins}}{592\text{ ft}^2}$$

$$= 0.285\text{ sabins}$$

and

$$R = \Sigma S_i\alpha_i/(1 - \alpha_{avg})$$

$$= \frac{168.96\text{ ft}^2\text{ sabins}}{1 - 0.285\text{ sabins}}$$

$$= 236.307\text{ ft}^2\text{ sabins or } 236\text{ ft}^2$$

and then

$$r_c = \sqrt{QR/16\pi}$$

$$= [2(236\text{ ft}^2)/16\pi]^{1/2}$$

$$r_c = 3.06\text{ ft}$$

Now compute the sound-pressure level at the critical distance.

$$SPL = SWL + 10\text{ Log }[(Q/4\pi r^2) + (4/R)] + 10.5 + T$$

$$= 110\text{ dB} + 10\text{ Log }[2/4\pi(3.06\text{ ft})^2] + [4/236\text{ ft}^2] + 10.5 + 0$$

$$= 110\text{ dB} + (-14.692) + 10.5$$

$$= 105.808\text{ dB (reference: } 0.00002\text{ N/m}^2)$$

Now compute the sound-pressure level somewhere in the inverse-square-law field, 0.7 feet.

$$SPL = SWL + 10 \text{ Log } [(Q/4\pi r^2) + (4/R)] + 10.5 + T$$

$$= 110 \text{ dB} + 10 \text{ Log } [(2/4\pi(0.7 \text{ ft})^2) + 4/236 \text{ ft}^2] + 10.5 + 0$$

$$= 110 \text{ dB} + (-4.663) + 10.5$$

$$= 115.837 \text{ dB (reference: } 0.00002 \text{ N/m}^2)$$

Now compute the sound-pressure level near the wall in the reverberant field at 7 feet.

$$SPL = SWL + 10 \text{ Log } [(Q/4\pi r^2) + (4/R)] + 10.5 + T$$

$$= 110 \text{ dB} + 10 \text{ Log } [(2/4\pi(7 \text{ ft})^2) + 4/236 \text{ ft}^2] + 10.5 + 0$$

$$= 110 \text{ dB} + (-16.947) + 10.5$$

$$= 103.553 \text{ dB (reference: } 0.00002 \text{ N/m}^2)$$

In the given room, then, the sound-pressure levels are 115.837 dB in the inverse-square-law field, 105.818 dB at the critical distance, and 103.553 dB near the wall.

Now change the characteristics of the room to see what happens to sound pressure levels in these three locations. Let the ceiling absorption coefficient remain at 0.95 sabins; let the floor absorption coefficient remain at 0.37 sabins; but change the walls from brick (absorption coefficient 0.03 sabins) to a material that has an absorption coefficient of 0.10 sabins. Then

$$\alpha_{avg} = \Sigma S_i \alpha_i / \Sigma S_i$$

$$= \frac{(12 \text{ ft} \times 10 \text{ ft}) \, 0.95 \text{ sabins} + (12 \text{ ft} \times 10 \text{ ft}) \, 0.37 \text{ sabins} + 2(12 \text{ ft} \times 8 \text{ ft}) \, 0.10 \text{ sabins} + 2(10 \text{ ft} \times 8 \text{ ft}) \, 0.10 \text{ sabins}}{592 \text{ ft}^2}$$

$$= \frac{193.6 \text{ ft}^2 \text{ sabins}}{592 \text{ ft}^2}$$

$$\alpha = 0.327 \text{ sabins}$$

and

$$R = \Sigma S_i \alpha_i / 1 - \alpha_{avg}$$

$$= \frac{193.6 \text{ ft}^2 \text{ sabins}}{1 - 0.327 \text{ sabins}}$$

$$= 287.667 \text{ ft}^2$$

At 0.7 ft in the inverse-square-law field (same distance as used above) the sound-pressure level in the more absorbent room would be

$$SPL = SWL + 10 \text{ Log } [(Q/4\pi r^2) + (4/R)] + 10.5 + T$$

$$= 110 \text{ dB} + 10 \text{ Log } [(2/4\pi(0.7 \text{ ft})^2) + (4/287.667 \text{ ft}^2)] + 10.5 + 0$$

$$= 110 \text{ dB} + (-4.702) + 10.5$$

$$= 115.798 \text{ dB (reference: } 0.00002 \text{ N/m}^2)$$

Recall that the sound-pressure level in the less absorbent room was 115.837 dB, a difference of only 0.039 dB.

At the critical distance (3.06 ft) the sound-pressure level in the more absorbent room would be

$$SPL = SWL + 10 \text{ Log } [(Q/4\pi r^2) + (4/R)] + 10.5 + T$$

$$= 110 \text{ dB} + 10 \text{ Log } [(2/4\pi(3.06 \text{ ft})^2) + (4/287.667 \text{ ft}^2)] + 10.5 + T$$

$$= 110 \text{ dB} + (-15.100) + 10.5$$

$$= 105.399 \text{ dB (reference: } 0.00002 \text{ N/m}^2)$$

compared to 105.808 dB in the less absorbent room, a difference of 0.409 dB.

Near the wall (7 feet) the sound-pressure level in the more absorbent room would be

$$SPL = SWL + 10 \text{ Log } [(Q/4\pi r^2) + (4/R)] + 10.5 + T$$

$$= 110 \text{ dB} + 10 \text{ Log } [(2/4\pi(7 \text{ ft})^2) + (4/287.667 \text{ ft}^2)] + 10.5 + 0$$

$$= 110 \text{ dB} + (-17.657) + 10.5$$

$$= 102.847 \text{ dB (reference: } 0.00002 \text{ N/m}^2)$$

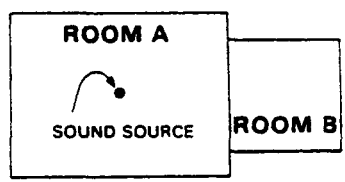
compared to 103.553 dB in the less absorbent room, this is a difference of 0.71 dB.

It can be seen from this example that making the room more absorbent will not affect the sound-pressure level in the inverse-square region or at the critical distance. The benefits of making the room more absorbent appear in the reverberant field. Making the walls more absorbent will not help the worker who is in the inverse-square-law region or at the critical distance; it will help only when in the reverberant field.

Sound in an Adjoining Room

So far, sound in only a single room has been discussed. This section deals with how sound travels from room to room. The situation would be something like that shown in Figure 4.3.9. In this figure, the sound source is in the larger room. The problem is to find the sound pressure level in the adjacent, smaller room (room B). The problem of concern is how sound travels through the wall that joins the secondary room (room B) and the primary room (room A).

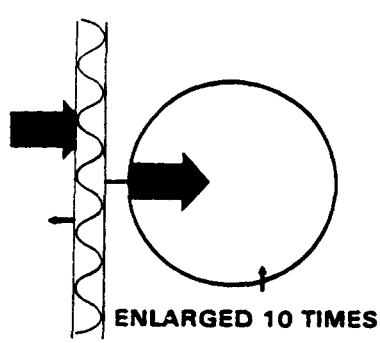
Figure 4.3.9



How does sound travel through a wall? If the wall is made of a highly absorbent material, a great deal of sound will go through the wall as depicted by Figure 4.3.10.

Figure 4.3.10

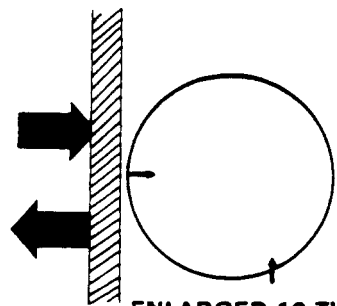
Sound travelling through a sound-absorbing wall.



If the wall or partition is solid, less sound goes through; and much more sound is deflected as depicted by Figure 4.3.11.

Figure 4.3.11

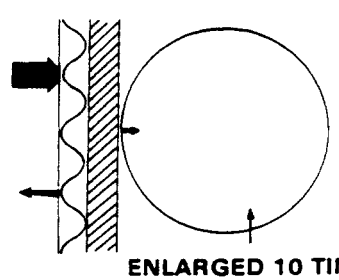
Sound travelling through a solid wall.



If a combination wall is built, a wall with highly absorbing material on one side and a hard or solid material on the other, little sound is reflected, and little sound goes through the wall as shown by Figure 4.3.12.

Figure 4.3.12

Sound travelling through a combination wall.



The amount of sound that is lost in the transmission through the wall depends upon the mass of the wall. Theoretically, the sound pressure on the source side of the wall or partition causes the partition to move in and out--vibrate. This motion causes a new sound wave on the opposite side of the partition. The movement of the partition or wall is dependent upon the mass of the material. A brick wall moves much less than a wood wall.

The transmission loss can be defined as the difference in the sound-pressure level on the source side compared to the sound-pressure level

on the opposite side of the wall; i.e., the transmission loss of a wall may be defined in terms of sound-pressure level reduction afforded by the wall. However, mathematically, the transmission loss (TL) is expressed as

$$TL = 10 \text{ Log}_{10}(\text{Energy transmitted/ Energy of incident})$$

Transmission loss is difficult to measure. Only a few laboratories in the United States are qualified to make the standard measures (ASTM E90-61T). However, there are tables available that give the transmission loss of standard materials. (See Table 4.3.1.) Note from the tables that the transmission loss is not the same for a given material at all frequencies. The transmission loss is dependent upon the frequency of concern.

Transmission Loss of Combined Materials

Most walls are not made of a single material. For example, walls are made of brick and glass, plaster and glass, plaster and wood, etc. When a wall is constructed of more than one material, the materials that make up the walls have different transmission losses. The transmission loss of a wall composed of a combination of materials can be computed as

$$TL_{combined} = 10 \text{ Log } \sum_{i=1}^{n} S_i - 10 \text{ Log } (\Sigma S \, 10^{TL_i/10})$$

where TL_i denotes the transmission loss of the ith material composing the wall.

Problem. A wall is 8 ft high and 12 ft wide and is made of 6-inch hollow concrete block (cinder aggregate) with no surface treatment. In the center of the wall is a 1/4 inch thick glass window, 20 inches x 40 inches. What is the transmission loss of the wall? Assume the frequency of concern is 512 Hz.

Solution. From Table 4.3.1, the 6-inch hollow concrete block has a transmission loss of 45 dB, and the glass has a transmission loss of 31 dB. Thus, the combined transmission loss would be

$$TL_{combined} = 10 \text{ Log } \sum_{i=1}^{n} S_i - 10 \text{ Log } (\Sigma S \, 10^{TL_i/10})$$

where

$$\sum_{i=1}^{n} S_i = 8 \text{ ft} \times 12 \text{ ft}$$

$$= 96 \text{ ft}^2$$

and the surface area of the concrete block is

$$S_{block} = 16 \text{ ft}^2 - (3.33 \text{ ft} \times 1.67 \text{ ft})$$

Table 4.3.1

Sound transmission loss of general building material and structures.

Item	Material or Structure	Weight Lbs/Ft^2	Loss in Decibels at Indicated Frequencies, Hz								
			128	192	256	384	512	768	1024	2048	4096
A	Doors										
1	Heavy wooden door, approx. 2-1/2" thick, special hardware, rubber gasket & sides, drop felt on bottom		23	26	26	28	29	30	26	33	33
2	Approx. same as above (No. 1)	12.5	30	30	30	29	24	25	26	37	36
3a	Wooden door, 2-5/8" thick, 3' x 7' with double frame construction, frames insulated with hair felt, surface formed of 1/4" hardwood panels, door hung in split frame with felt insert & mounted in 12" brick wall. Two tubular gaskets gave a double seal around both sides & top of door, two drop felts at bottom	6.8	29	33	33	32	36	34	34	41	40
4a	Same as panel A-3, except door edges plastered to frame on both sides		32	38	38	35	39	38	42	49	53
5a	Door, steel clad (well sealed to door casing and threshold)		42	47	51	48	48	45	46	48	45
6a	Door, flush hollow core (well sealed to door casing and threshold)		14	21	27	24	25	25	26	29	31
7	Door, oak, solid 1-3/4", with cracks as ordinarily hung		12		15		20		22	16	
8a	Door, 3" x 30" x 84" wooden door, special soundproof construction; sponge rubber gasket around sides and top, approx. 1/2" square cross section, chamfered on hinge side; and a sponge rubber drop closure at bottom of door	7	31	27	32	30	33	31	29	37	41
B	Glass										
1a	Glass, 1/8" double strength, 40" x 20"		15		26		27		31	33	29
2a	Glass, 1/4" plate, 40" x 20"		25		33		31		34	34	32
	Walls - Homogeneous										
1	Single sheet of .025" aluminum, 40" x 21-1/2" (Approx. 22 ga)	0.35			18		13		18	23	25[b]
2	Single sheet of .03" galvanized iron, 40" x 21-1/2" (Approx. 22 ga)	1.2			25		20		29	35	32
3	Single sheet of lead 1/8" thick	6.2			31.0		27.2		37.5	43.8	32.6[b]
4	Single sheet of lead 1/16" thick	3.9			31.8		33.2		32.0	32.1	32.5[b]
5	PF fiberglas, 6 lb./ft.^{3} density, 1" thick	0.5	5	5	5	5	5	4.5	4.5	4	3.5
6	PF fiberglas, 6 lb./ft.^{3} density, 2" thick	1.0	8	8	8	8	7.5	7.5	7	7	6.5
7	PF fiberglas, 6 lb//ft.^{3} density, 3" thick	1.5	10	10	10	10	10	10	10	10	10
8	PF fiberglas, 6 lb./ft.^{3} density, 4"thick	2.0	12	12	12.25	12.5	12.5	12.5	12.5	13	13
9	Glass brick partition, 3-3/4" x 4-7/8" x 8" thick		30.2	36.2	34.7	39.4	40.5	45.1	48.6	49.0	43.4
10	Concrete, reinforced, 4"		37	33	36	44	45	50	52	60	67
11	Concrete, reinforced, 8"		42	38	41	49	50	55	57	65	72
12	Cinder block, 4" solid	24.4	22	25	29	31	31	32	34	40	41
13	Cinder block, 8" solid	40.2	17	22	24	35	40	45.5	51.5	58.5	62.5

[a]Results obtained at 125, 175, 250, 350, 500, 700, 1000, 2000, 4000 cps

[b]Results obtained at 3100 cps instead of 4096 cps.

Table 4.3.1 (continued)

Item	Material or Structure	Weight Lbs/Ft^2	Loss in Decibels at Indicated Frequencies, Hz								
			128	192	256	384	512	768	1024	2048	4096
D	Walls - Nonhomogeneous										
1	2 x 4 studs, 1/2" insulating board on both sides (16" ctrs)	3.8	15.9	19.0	21.9	32.4	28.1	33.3	37.9	49.7	51.8
2	2 x 4 studs, 3/4" insulating board on both sides (16" ctrs)	4.3	20.7	17.8	21.4	26.6	20.7	32.1	37.6	49.3	52.7
3	2 x 4 studs, 3/8" Gypsum board lath & 1/2" gypsum plaster on both sides (16" ctrs)	15.0	33.4	23.9	24.5	30.2	28.3	37.7	35.5	41.7	58.8
4	2 x 4 studs, 1/2" insulating board lath & 1/2" gypsum plaster on both sides (16" ctrs)	12.6	28.3	27.3	30.9	37.8	40.7	44.4	46.3	46.7	65.6
5	2 x 4 studs, 1/2" dense wood fiberboard both sides (16" ctrs) joints at studs	3.8	16	19	22	32	28	33	38	50	52
6	2 x 4 studs, 2 sets staggered (16" cts with studs of one set 8" on ctrs and projecting 2" on ctrs fromother set) 1/2" wood fiberboard both sides, joints filled	4.9		34[c]	30		28		42	59	60[c]
7	Concrete block - 4" hollow, no surface treatment)		27	29	32	35	37	42	45	48	48
8	Concrete block - 4" hollow, one coat resin - emulsion paint)		30	33	34	36	41	45	50	55	53
9	Concrete block - 4" hollow, one coat cement base paint) Cinder		37	40	43	45	46	49	54	56	55
10	Concrete block - 6" hollow, no surface treatment) Aggregate		28	34	36	41	45	48	51	52	47
11	Concrete block - 8" hollow, no surface treatment)		18	24	28	34	37	39	40	42	40
12	Concrete block - 8" hollow, one coat cement base paint)		30	36	40	44	46	48	51	50	41
13	Concrete block - 8" hollow, filled with vermiculite insulators)		20	29	33	36	38	38	40	45	47
14	Concrete block - 4" hollow, no surface treatment)		21	26	28	31	35	38	41	44	43
15	Concrete block - 4" hollow, one coat resin-emulsion paint) Expanded		26	30	32	34	37	42	43	46	44
16	Concrete block - 4" hollow, two coats resin-emulsion paint) Shale		24	31	33	35	38	42	44	47	44
17	Concrete block - 4" hollow, one coat cement-base paint) Aggregate		23	30	35	38	42	43	44	48	43
18	Concrete block - 4" hollow, two coats cement-base paint)		34	38	40	42	45	47	49	51	46
19	Concrete block - 6" hollow, no surface treatment)		22	27	32	36	40	43	46	45	43
20	Concrete block - 4" hollow, no surface treatment)		30	36	39	41	43	44	47	54	50
21	Concrete block - 4" hollow, one coat cement base paint on face)		30	36	39	41	43	44	47	54	49
22	Concrete block - 6" hollow, no surface treatment)		37	46	50	50	50	53	56	56	46
23	Concrete block - 6" hollow, one coat resin-emulsion paint) Dense *each face*) *Aggre-*		37	50	54	52	53	55	57	56	46
24	Concrete block - 8" hollow, no surface treatment) gate		40	47	53	54	54	56	58	58	50
25	Concrete block - 8" hollow, two coats resin-emulsion paint each face)		38	50	54	54	55	58	60	58	49
26	Fluted steel, 18 gauge, stiffened at edges by 2 x 4" wood strips, joints sealed	4.4	30	20	20	21	22	17	30	28	31
27	Fluted steel, 18 gauge, 1-1/2" mineral wool - flat steel sheet, 18 gauge panel stiffened by 2 x 8" wood beam, joints caulked	7.8	36	30	25	36	37	42	46	44	44
28	Corrugated asbestos - bolted to 2 x 8" stiffening beam, braced at top and bottom by asphalt strips, joints sealed	7.0	33	29	31	34	33	33	33	42	39
29	Same as No. 29 except backed by 1-3/16" uncorrugated board, composed of 15/16" organic matl. covered both sides by 1/8" asbestos fiberboard, joints closed by 1 x 1" furing, joints sealed	10.4	40	36	33	38	40	43	46	45	42
30	Tile, hollow - 4 x 8 x 16" - surface untreated	15.5	8	8	5	7	9	12	14	18	17
31	Tile, hollow - 4 x 8 x 16" - one surface plastered	20.4	31	27	27	36	35	33	36	40	47

[c]Results obtained at 165 & 3100 cps instead of 192 & 4096 cps

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$$= 95 \text{ ft}^2 - 5.561 \text{ ft}^2$$

$$= 90.439 \text{ ft}^2$$

and the surface area of the glass is

$$S_{glass} = (3.33 \text{ ft} \times 1.67 \text{ ft})$$

$$= 5.561 \text{ ft}^2$$

and the combined transmission loss would be

$$TL_{combined} = 10 \text{ Log } \sum_{i=1}^{n} S_i - 10 \text{ Log } (\Sigma S \, 10^{TL_i/10})$$

$$= 10 \text{ Log } 96 \text{ ft}^2 - 10 \text{ Log } [90.439 \text{ ft}^2 \times 10^{-45 \text{ dB}/10} +$$

$$5.561 \text{ ft}^2 \times 10^{-31 \text{ dB}/10}$$

$$= 19.823 - (-21.38)$$

$$= 41.203 \text{ dB}$$

Possible leaks are of concern when a wall is composed of different kinds of materials. A tiny portion of the wall or partition constructed of a material of low transmission loss tends to nullify the effect of the remainder of the wall. Consider the transmission loss of a 6-inch concrete wall 10 feet high and 8 feet wide. This structure would have a transmission loss of 45 dB in the mid frequencies (512 Hz). But suppose that a mason did not realize the importance of small cracks in a noise-control structure and carelessly left an opening of just one-eighth of an inch wide along the top of the wall. The area of this small crack would then be 0.083 square feet while the area of the block would be 79.917 square feet. The combined transmission loss of the concrete wall with the one-eighth inch crack at the top will only have an effective transmission loss of 29.679 dB; this is substantially less than would be expected for the concrete wall (45 dB). One must always be alert for similar situations in construction for the purpose of noise control. Leaks around doors are extremely hard to eliminate unless great care is exercised. Poorly fitted panels and joints are also frequently sources of trouble. The point is that leaks are going to reduce the transmission loss. These leaks will cause more noise to go into the adjoining room.

Transmission loss is the amount of sound lost as it passes through a wall. The higher the transmission loss, the more sound that is lost.

Secondary Room

In the last section, transmission loss was discussed. When sound travels through a wall, the wall begins to vibrate and creates sound on the opposite side of the wall. If the adjoining room is small, the energy--the sound

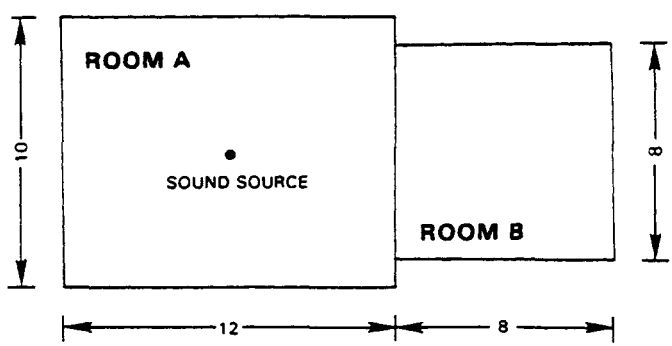
energy that gets through the wall--will bounce off the walls in the smaller room. This reverberant sound must also be considered if the sound-pressure level in the adjoining room is to be computed. Thus, if it is desired to compute the sound-pressure level in the secondary room as a result of the sound source in the primary room, it is necessary to consider the reverberant field in the secondary room.

The amount of reverberant sound in the secondary room depends upon the absorption coefficients of the walls, floor, and ceiling (the room constant) in the secondary room. This characteristic of the secondary room is taken into consideration in the computation of the noise reduction. Noise reduction is defined as the difference in the sound-pressure level from the source side of the wall to the opposite side of the wall when there is going to be reverberant buildup in the smaller room. The general formula for noise reduction (NR) is given as follows:

$$NR = TL - 10 \log_{10} [(1/4) + (S_w/R)]$$

where NR denotes the noise reduction--the difference in sound-pressure levels on the opposite sides of the walls; TL denotes the transmission loss through the partition or wall; S_w indicates the surface area of the wall between the two rooms; and R denotes the room constant in the secondary room.

Problem. Assume the following layout:



Both Rooms A and B are 8 feet high. In room A, let $\alpha_{avg} = 0.285$ sabins and let R = 237 ft^2. In room B, let the walls, ceiling, and floor all have an absorption coefficient of 0.07 sabins. Further, let the transmission loss of the adjoining wall (8 ft by 8 ft) be 45 dB. What is the noise reduction of the wall, considering the reverberant buildup in the smaller room?

Solution. The noise reduction can be computed from

$$NR = TL - 10 \log_{10} [(1/4) + (S_w/R)]$$

where $S_w = 64\ ft^2$ and the room constant in the smaller room is given as

$$R = \alpha \Sigma S_i / (1 - \alpha_{avg})$$

where

$$\alpha = \Sigma S_i \alpha_i / \Sigma S_i$$

$$= \frac{6(8\ ft \times 8\ ft)0.08\ sabins}{6(8 \times 8)}$$

$$= 0.08\ sabins$$

and thus

$$R = \frac{30.720\ ft^2\ sabins}{1\ sabin - 0.08\ sabins}$$

$$= 33.391\ ft^2$$

Thus

$$NR = TL - 10\ Log_{10}\ [(1/4) + (S_w/R)]$$

$$= 45\ dB - 10\ Log\ [(1/4) + (64\ ft^2/33.391\ ft^2)]$$

$$= 45\ dB - 3.358\ dB$$

$$= 41.642\ dB$$

Notice that the noise reduction is smaller than the transmission loss because the smaller room has a log average absorption coefficient, causing an excessive buildup of reverberant sound.

Given the noise reduction, the sound-pressure level in the smaller room can be computed if the sound-pressure level at the adjoining wall in the larger room is known. That is,

$$SPL_{in\ smaller\ room} = SPL_{at\ wall\ of\ larger\ room} - NR$$

For example, if the larger room has an $R = 237\ ft^2$ and a sound source with a sound power level of 120 dB and $Q = 2$, then the sound-pressure level at the wall in the larger room (an average of 6.86 feet from the center of the room) would be

$$SPL = SWL + 10\ Log\ [(Q/4\pi r^2) + (4/R)] + 10.5$$

$$= 120\ dB + 10\ Log\ [(2/4\pi(6.86\ ft)^2) + (4/237\ ft^2)] + 10.5$$

$$= 120\ dB + (-16.934) + 10.5$$

$$= 113.566\ dB\ (reference: 0.00002\ N/m^2)$$

and sound pressure level in the adjoining room would be calculated as

$$SPL_{\text{in smaller room}} = SPL_{\text{at wall in larger room}} - NR$$

$$= 113.566 \text{ dB} - 41.642 \text{ dB}$$

$$= 71.924 \text{ dB (reference: } 0.00002 \text{ N/m}^2\text{)}$$

Summary

The following principles were discussed in this chapter:

1. Sound in a room will be reflected off the walls making sound behave differently than in the free-field condition.

2. There are three fields of sound in a room: the near field (sound near and around the source), the field that follows the inverse-square law, and the reverberant field (sound near the walls of the room). The predominant sound in the inverse-square-law field is the sound generated by the source. The predominant sound in the reverberant field is sound bouncing (reflecting) off the walls.

3. The sound-pressure level in the inverse-square-law field is affected by the direction and distance from the sound source. The sound-pressure level in the reverberant field will become constant and is not affected by the distance from the sound source.

4. The sound-pressure level in the reverberant field is affected by the absorption ability of the walls, ceiling, and floor. A more absorbent material reflects less sound back into the room. A less absorbent material reflects more sound back into the room.

5. The absorption coefficient (α) indicates the absorption ability of the material. A material that has an absorption coefficient of 1 ($\alpha = 1$) is a perfectly absorbing material (such as open air). A very hard surface, such as marble, has an absorption coefficient equal to 0 ($\alpha = 0$). Inthis case no sound is absorbed; it is all reflected back into the room.

6. The absorption coefficient of a material is not the same value at all frequencies.

7. The average absorption coefficient of a room is computed by

$$\alpha_{avg} = \frac{\sum_{i=1}^{n} S_i \alpha_i}{\sum_{i=1}^{n} S_i}$$

where S_i denotes the surface area composed of the ith material and α_i denotes the absorption characteristics of the ith material. α is measured in sabin units.

8. The room constant (R) is a measure of the ability of a room to absorb sound. It is a function of the average absorption coefficient.

$$R = \frac{\alpha_{avg} \sum_{i=1}^{n} S_i}{1 - \alpha_{avg}}$$

R is in square feet units.

9. The average absorption coefficient (and therefore the room constant) affects the sound-pressure level in various parts of the room. If the average absorption coefficient equals 1 (a free-field condition), then

$$SPL_{(r)} = SPL_{(1)} - 20 \text{ Log } (r/r_1)$$

where $SPL_{(r)}$ denotes the sound-pressure level at distance r; $SPL_{(1)}$ denotes a known sound pressure at some distance r_1; and r and r_1 denote the distances in feet corresponding to $SPL_{(r)}$ and $SPL_{(1)}$. This relationship will not hold, however, if the materials composing the walls, floor, and ceiling are less than perfectly absorbent. There are tables to predict the sound-pressure level in various parts of the room if the walls, ceiling, and floor are composed of materials with less than perfectly absorbing characteristics.

10. The room constant (R) also affects the relationship between sound-power level and sound-pressure level.

$$SPL = SWL + 10 \text{ Log } [(Q/4\pi r^2) + (4/R)] + 10.5 + T$$

where SPL denotes sound-pressure level (reference: 0.00002 N/m^2), SWL denotes sound-power level (reference: 10^{-12} watts), Q denotes the directivity factor, r denotes distance from the sound source in feet, R denotes the room constant in square feet, and T denotes the correction for nonstandard atmospheric conditions.

11. The critical distance (r_c) is the point where the inverse square law field ends and the reverberant field begins.

$$r_c = \sqrt{QR/16\pi}$$

where Q is the directivity factor and R is the room constant. Alternatively, r_c can be expressed as

$$r_c = \sqrt{(Q\Sigma S\alpha_{avg})/16\pi(1-\alpha_{avg})}$$

where S denotes the surface area in feet of the material and α_{avg} denotes the average absorption coefficient.

12. The critical distance is important because most workers are within the critical distance zone. Further, it can be shown that making the floor, ceiling, and walls more absorbent does not significantly affect the sound-pressure level at the critical distance. However, making the room more absorbent will reduce the sound-pressure level near the walls.

13. Sound will travel from one room to another. Theoretically, the sound-pressure level on the source side of a wall will cause the wall to vibrate. This motion causes a new sound wave on the opposite side of the wall.

14. The transmission loss (TL) of a wall can be defined in terms of the sound-pressure level reduction afforded by the wall. Mathematically,

$$TL = 10 \text{ Log } (\text{Energy transmitted} / \text{Energy of incident})$$

Transmission loss of materials is usually published in table form.

15. If an adjoining wall is composed of various material, the combined transmission loss can be computed as follows:

$$TL_{combined} = 10 \text{ Log } \sum_{i=1}^{n} S_i - 10 \text{ Log } (\Sigma S \ 10^{TL_i/10})$$

where S_i denotes the surface area (in square feet) of the ith material composing the wall and TL_i denotes the transmission loss of the ith material composing the wall.

16. If there are leaks in the adjoining wall (air spaces around windows, doors, etc.), the effective transmission loss is considerably reduced.

17. If the room adjoining the sound source room is small, then the sound transmitted from the adjoining wall will cause a reverberant buildup in the small room, thus possibly increasing the sound-pressure level in the smaller room.

18. Noise reduction is defined as the difference in the sound-pressure level from the source side of the wall to the opposite side of the wall. When there is going to be reverberant buildup in the smaller room, the noise reduction is computed as follows:

$$NR = TL - 10 \text{ Log}_{10} [(1/4) + (S_w/R)]$$

where TL denotes the transmission loss of the adjoining wall, S_w denotes the surface area of the adjoining wall in square feet, and R denotes the room constant of the adjoining room.

19. The amount of noise reduction is related to the room constant (and thus to the average absorption coefficient) of the adjoining room. If the adjoining room has a small average absorption coefficient, the noise reduction achieved will be 5 to 6 dB less than transmission loss. The noise reduction will be less than the transmission loss due to the extreme

buildup in the adjoining room. If the average room absorption coefficient is average, the noise reduction achieved will be 1 to 2 dB greater than the transmission loss. If the adjoining room has a very high average absorption coefficient, the noise reduction achieved will be about 5 dB greater than the transmission loss because the sound transmitted will be absorbed by the wall, ceiling, and floors of the adjoining room.

20. The sound-pressure level in the adjoining room can be computed as follows:

a. Compute the average distance the wall is from the source in the sound source room using

$$r_{\substack{\text{average}\\ \text{to wall}}} = \sqrt{\Sigma S/4\pi}$$

where S denotes the surface area in the source room.

b. Compute the average absorption coefficient in the source room.

$$\alpha_{avg} = \frac{\sum_{i=1}^{n} S_i \alpha_i}{\sum_{i=1}^{n} S_i}$$

where S_i denotes the surface area of the ith building material and $_i$ denotes the absorption coefficient of the ith building material.

c. Compute the room constant of the source room.

$$R = \frac{\alpha_{avg} \sum_{i=1}^{n} S_i}{1 - \alpha_{avg}}$$

d. Compute the sound-pressure level at the wall in the source room using

$$SPL = SWL + 10 \text{ Log } [(Q/4\pi r^2) + (4/R)] + 10.5 + T$$

where r becomes the average distance to the wall (step a) and R becomes the room constant of the source room (step c).

e. Compute the transmission loss of the material forming the adjacent wall (if wall is composed of many materials, using the following formula):

$$TL_c = 10 \text{ Log } \Sigma S_i - 10 \text{ Log } (\Sigma S_i \ 10^{-TL_i/10})$$

f. Compute the surface area of the adjoining wall (S_w).

g. Compute the average absorption coefficient in the adjoining room.

$$\alpha_{avg} = \frac{\sum_{i=1}^{n} S_i \alpha_i}{\sum_{i=1}^{n} S_i}$$

and then compute the room constant in the adjoining room.

$$R = \frac{\alpha_{avg} \sum_{i=1}^{n} S_i}{1 - \alpha_{avg}}$$

h. Compute the noise reduction of the wall by

$$NR = TL - 10 \text{ Log}_{10} [(1/4) + (S_w/R)]$$

where TL is the transmission loss computed in step e, S_w is the surface area of the adjoining wall (step f), and R denotes the room constant in the adjoining room (step g).

i. Compute the sound pressure level in the adjoining room by

SPL adjoining room = SPL at wall in source room - NR

where SPL (at wall in source room) is given by step d and NR is given by step h.